🔗 Part of the ATRAPA database project

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Validating OpenStreetMap for Detecting Cycling Infrastructure Change: A Pilot Study in Barcelona

Introduction

- Context: Urban transformations (e.g. new bike lanes, pedestrian areas) can reshape mobility and health, but studying their effects requires reliable historical data.
- Problem: Standardised datasets of infrastructure change are rarely available across cities and years.
- Potential solution: Volunteered Geographic Information, especially OpenStreetMap (OSM), provides open and historical data on infrastructure, offering a promising way to track changes over time. However, the reliability of OSM data for detecting infrastructure change is uncertain and requires systematic validation.
- Aim: Assess how well OSM detects cycling infrastructure (CI) changes (additions/removals), by comparing OSM-derived changes with GSV imagery, and by estimating precision, recall, false negatives, and error-adjusted rates of change.
- Contribution: Using Barcelona as a pilot case, this study provides one of the first systematic validations of OSM for detecting changes in CI, introduces an error-adjusted approach to estimate true additions and removals, and delivers city-level diagnostics to inform cross-city comparisons and practical monitoring of urban transformations.

Data and Methods

Data sources

OpenStreetMap (OSM)

Google Street View (GSV)

Methods

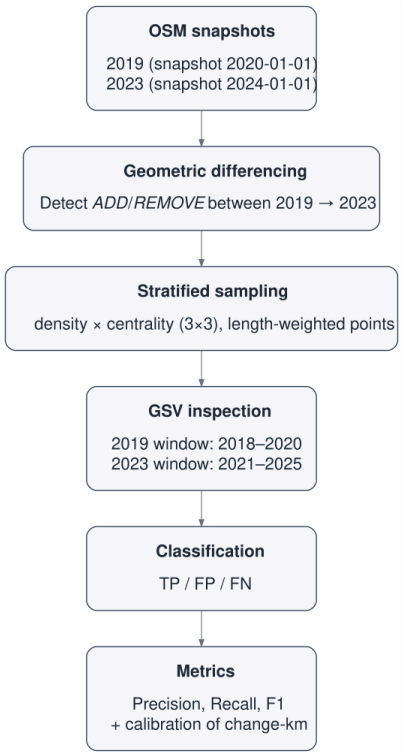

Figure 1: Analytical workflow showing the main steps for detecting and validating cycling-infrastructure changes in Barcelona (2019–2023).

OSM data extraction and preprocessing

- Extraction: Download OSM linework for both years (snapshot dates: 2020-01-01 for 2019 and 2024-01-01 for 2023).
- CI selector:
  - Keep segments with: `highway=cycleway` • `cycleway=*` or `cycleway:left/right/both=*` (lane/track/opposite/separate) • `bicycle_road=yes` • designated `path/footway/track` for bicycles.
  - Normalize geometries; metric CRS; min-length filter; tolerance buffer to reduce sliver noise.
- Non-CI (general network): Base roads excluding all CI tags (strict complement).

Change detection

- ADDED: present in 2023, absent in 2019.
- REMOVED: present in 2019, absent in 2023.
- Computed via buffered geometric differencing, dropping tiny fragments.

Sampling design

- Stratified tracts: Pre-selected census tracts by centrality × density (3×3).
- In sampled tracts, draw length-weighted points for ADD, REMOVE, and GENERAL (2023 non-CI network).
- Validation point = midpoint of the sampled segment.

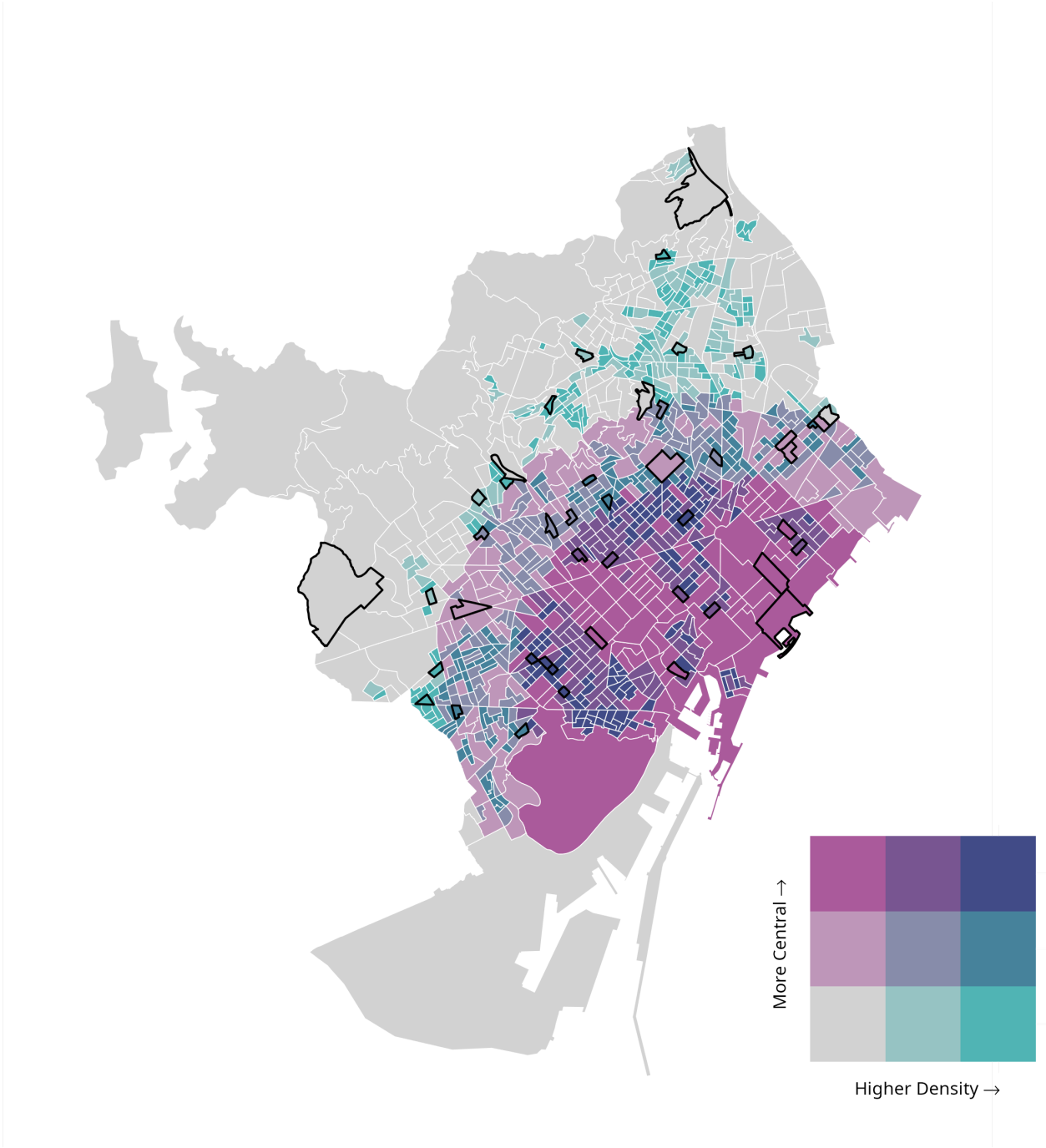

Figure 2: Stratified sampling bivariate map (density × centrality, 3×3) with selected validation tracts; inset shows the bivariate legend.

GSV inspection

- Tolerance windows. 2019 condition: 2018–2020; 2023 condition: 2021–2025.
- Procedure. At each sampled point, inspect both windows; if either window is not verifiable, exclude the point from accuracy stats.
- Variables recorded (per year). verifiable_[year] (Y/N), presence_[year] (Y/N/NA), notes_[year] (free text).

Classification

- OSM-flagged ADD. TP if CI absent in 2019 and present in 2023; otherwise FP.
- OSM-flagged REMOVE. TP if CI present in 2019 and absent in 2023; otherwise FP.
- GENERAL points. Used to detect FN (true changes that OSM did not flag).
- Usable points. Only points with both years verifiable enter precision/recall.

Figure 3: Spatial distribution of validation points across the nine density–centrality strata in Barcelona (2019–2023).

Performance metrics

- Precision = TP / (TP + FP)
- Recall = TP / (TP + FN)
- F1 = 2·Precision·Recall / (Precision + Recall)
- Notes. Precision is computed on OSM-flagged points (ADD/REMOVE); FN comes from GENERAL points. Report metrics by class and pooled; add simple CIs if space allows.

Results

Raw OSM change estimates

The OSM cycling network expanded from 214.1 km in 2019 to 263.9 km in 2023 (+49.7 km, ≈23%). Segment differencing suggests 63.7 km were added and 22.3 km removed (Table 1). Additions were concentrated in dense, central tracts—especially D3_C3 (36.1% of added km)—while removals were less frequent and clustered mainly in D1_C3 (61.0%) (Table 2).

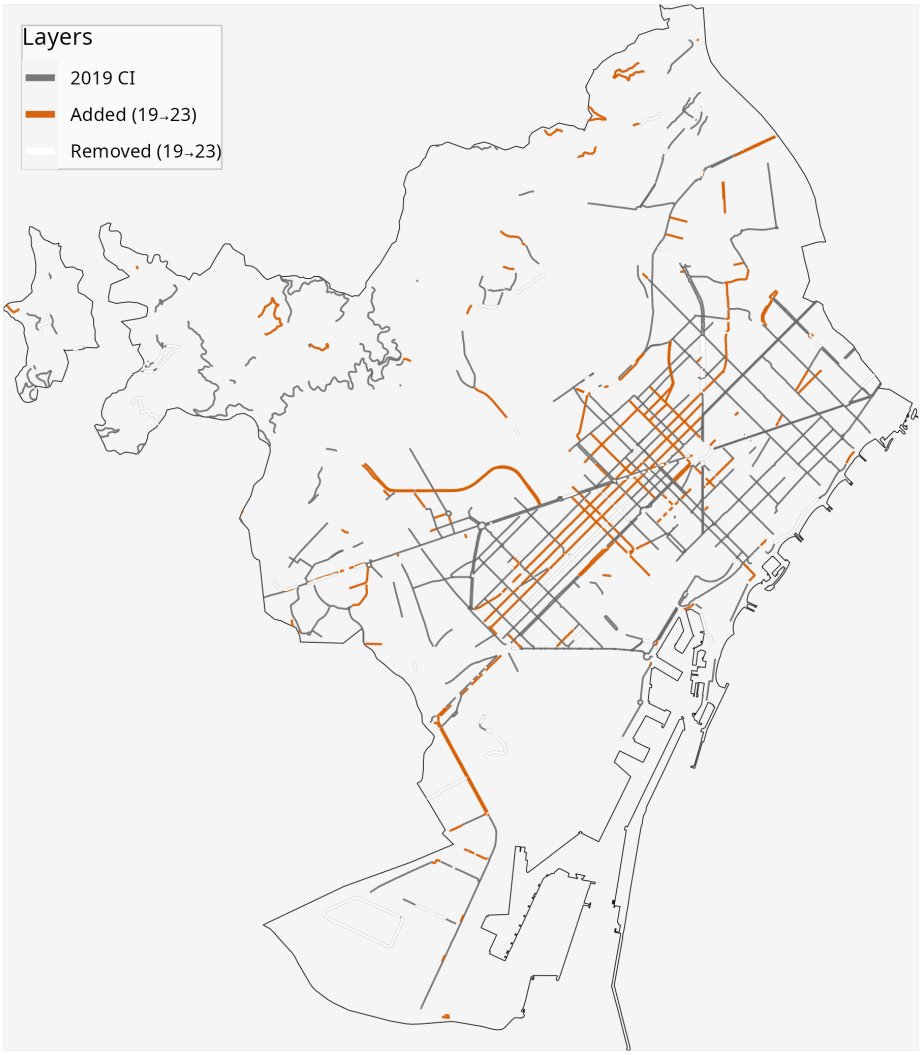

Figure 4: Cycling-infrastructure changes detected in OpenStreetMap between 2019 and 2023 in Barcelona. Additions and removals are shown by type and spatial distribution.

Table 1: Consistency between yearly totals and differencing estimates (2019–2023, Barcelona)

Metric	Value
Total 2019 (km)	214.1
Total 2023 (km)	263.9
Net growth (km)	49.7
Added (km)	63.4
Removed (km)	22.3
Added – Removed (km)	41.1
Gap: (Added–Removed) – Net	-8.6

Table 2: OSM-estimated additions and removals (2019–2023) by density × centrality stratum

stratum	Added_km	Removed_km	Added_pct	Removed_pct
D1_C1	0.0	0.0	0.0%	3.1%
D1_C2	0.4	0.0	22.1%	4.9%
D1_C3	0.3	0.5	13.3%	61.0%
D2_C1	0.0	0.1	0.0%	11.1%
D2_C2	0.1	0.2	5.2%	19.8%
D2_C3	0.1	0.0	4.3%	0.0%
D3_C1	0.2	0.0	9.8%	0.0%
D3_C2	0.2	0.0	9.3%	0.0%
D3_C3	0.7	0.0	36.1%	0.0%

Validation results

Validation covered 44 points across all nine density–centrality strata (largest D3_C3: 11/44). Among OSM-flagged changes with verifiable imagery (n=20; 15 ADD, 5 REMOVE), ADD performed well (precision 0.87 [0.62–0.96]; recall 1.00 [0.77–1.00]; F1 0.93), whereas REMOVE was weak (precision 0.20 [0.04–0.62]; recall 0.50 [0.09–0.91]; F1 0.29); we found one false negative (missed removal) and none for ADD. Pooled over classes: precision 0.70 [0.48–0.85], recall 0.93 [0.70–0.99], F1 0.80.

Table 3: Validation points by class and stratum (usable points)

stratum	REMOVE	ADD	GENERAL	Total
D1_C1	1	0	0	1
D1_C2	1	3	4	8
D1_C3	1	2	3	6
D2_C1	1	0	0	1
D2_C2	1	1	2	4
D2_C3	0	1	2	3
D3_C1	0	1	2	3
D3_C2	0	2	2	4
D3_C3	0	5	5	10
TOTAL	5	15	20	40

Table 4: Validation performance (usable points only)

Class	n (usable)	TP	FP	FN (source)	Precision	Precision (95% CI)	Recall	Recall (95% CI)	F1
ADD	15	13	2	from GENERAL	0.87	0.87 [0.62–0.96]	1.00	1.00 [0.77–1.00]	0.93
REMOVE	5	1	4	from GENERAL	0.20	0.20 [0.04–0.62]	0.50	0.50 [0.09–0.91]	0.29
Pooled	20	14	6	from GENERAL	0.70	0.70 [0.48–0.85]	0.93	0.93 [0.70–0.99]	0.80

Supplements

S1. Validation workbook

Data: Download the validation workbook (Excel)

S2. Interactive stratification map (density × centrality, 3×3) with sampled tracts

S3. Interactive validation points map (with GSV links)

S4. Interactive infrastructure change map (2019 → 2023)